

DATA ANALYSIS & USER EXPERIENCE REPORT

Deep Dive into Market Trends and Consumer Behavior

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Languages: AR | EN | TR | FR

23%

Electronics Revenue Share

3/10

Avg. Customer Rating

39%

Direct Deposit Usage

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Executive Summary

This report conducts a comprehensive study of the items sold in the store, analyzes customer behavior based on the most frequent purchase patterns, and evaluates customer satisfaction through reviews. It provides data-driven solutions to address existing challenges and optimize store performance.

Key Findings

Category Performance

The **Electronics** category is the primary revenue driver, accounting for more than **23%** of total turnover. This highlights its strategic importance for the store's financial growth.

Customer Behavior

Direct Deposit is the most preferred payment method, accounting for **39%** of all transactions — reflecting high customer trust in this channel.

Customer Satisfaction

The average customer rating is **3 / 10**, indicating a significant decline in overall satisfaction. This necessitates a deep-dive into root causes to urgently address technical or operational issues and improve the user experience.

Statistical Insight

The T-test showed no significant difference between payment methods and ratings ($P = 0.7$), thus ruling out a problem with the payment gateway.

Data Cleaning & Processing

- **205 rows and 7 columns:**Original dataset
- **5 duplicate rows removed to ensure data uniqueness :**Deduplication
- **Missing values filled using the Mean:**Price imputation
- **Missing values filled using the Median**Ratings imputation
- **Replaced with 0 and cast to Float for numerical analysis**FREE entries

- **Missing values imputed using the Mode:**Payment method
- **Missing Not At Random values handled rigorously to ensure unbiased results:**MNAR handling

Statistical Standards

Analysis was conducted at a 95% Confidence Level, utilizing Student's T-test and P-value interpretation to validate hypothesis significance. Data integrity was ensured through IQR outlier detection and descriptive statistical validation to provide unbiased, actionable insights.

Technology Stack

Tool	Purpose
Pandas & NumPy	Table processing and numerical computation
Matplotlib & Seaborn	Professional data visualizations and charts
SciPy	Statistical testing (Student's T-test)

Data Visualizations

01 Revenue by Category

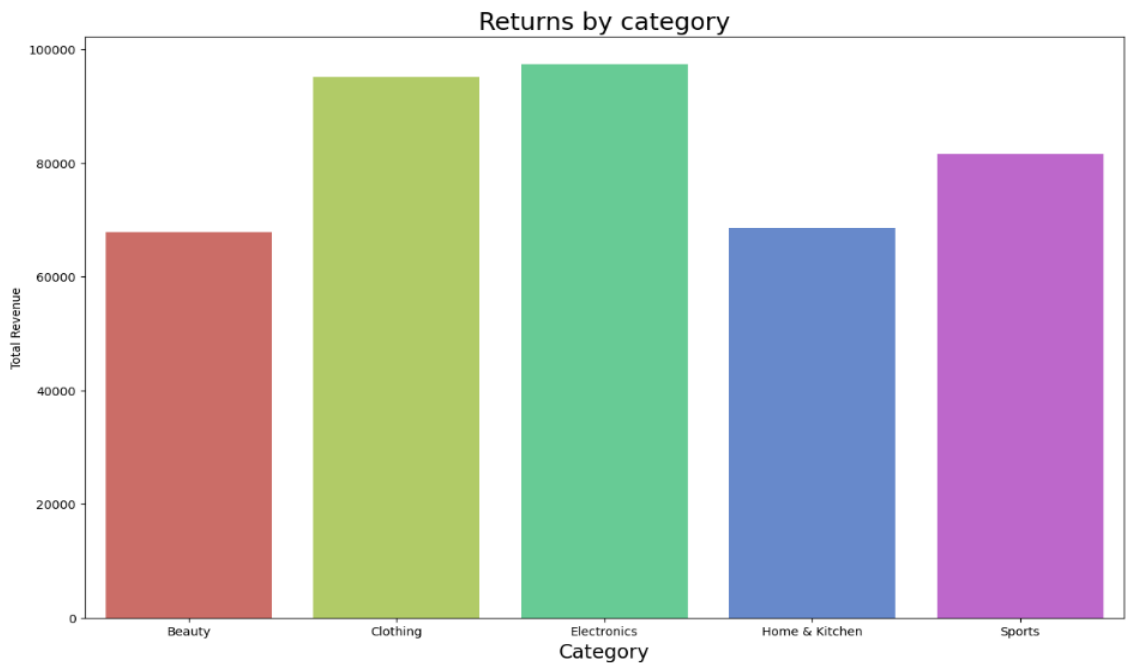


Figure 1 — Total revenue contribution per product categ

Observation : The Electronics category contributes over 23% of total revenue, outperforming all other categories and highlighting strong market demand.

Action : it is recommended to prioritize this category and increase investment to further drive revenue growth.

02 Distribution of Customer Satisfaction Levels

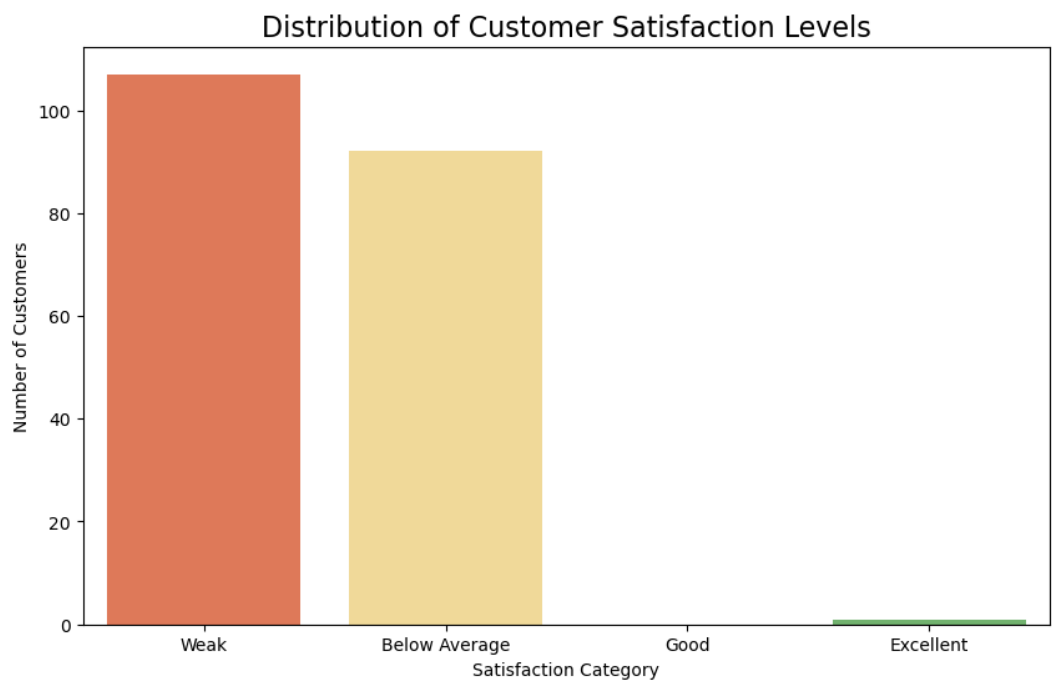


Figure 2 — Customer satisfaction breakdown across all categories

Observation : The **3/10** average reveals a critical satisfaction gap

Action : necessitating an immediate quality audit

03 Payment Methods Distribution

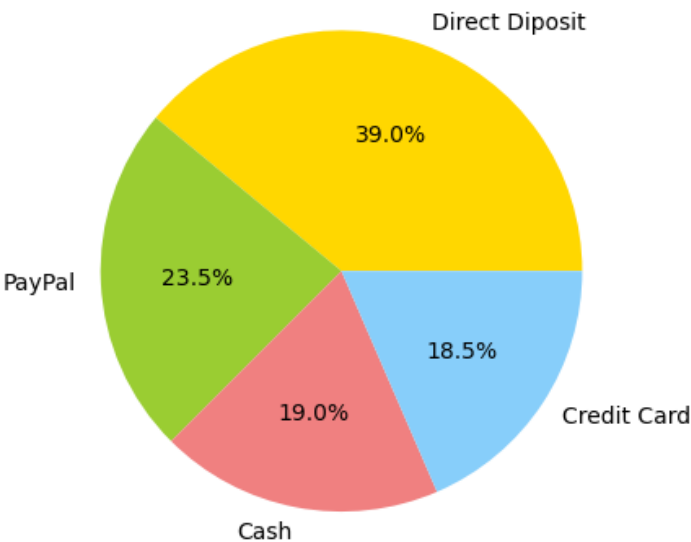


Figure 3 — Share of transactions by payment method

Observation : Direct Deposit is the top choice (**39%**) of all transactions. This reflects strong customer confidence in this payment method, positioning it as a key channel within the overall transaction strategy.

Action : Focus on this payment method

Stratigic Recomondation

Focus Area	Recommended Action	Priority
Customer Satisfaction	Send automated post-purchase surveys	● High
Payment Loyalty	Offer exclusive discounts for Direct Deposit users	□ Medium
Inventory	Increase electronics budget & audit supplier quality	● High

Value Added Summary

Raw data was transformed into actionable insights. By addressing gaps in customer experience (ratings) and boosting winning departments (electronics), the store can achieve sustainable growth and increase return on investment (ROI).

Contact & Sign-off

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Thank you for your trust — we look forward to helping you turn more data into successes.